

Edgar Rodríguez de la Calle

Unity Developer · Interactive Applications & Games

Barcelona · +34 615 46 49 41 · edirc98@gmail.com · [linkedin/edgarrodriguezdelacalle](https://www.linkedin.com/in/edgarrodriguezdelacalle) · [github/edirc98](https://github.com/edirc98)

PROFILE

Unity Developer with 5+ years shipping interactive applications across simulation, clinical research, and game development. Experienced in the full development lifecycle — from architecture and gameplay systems to integration, testing, and delivery. Focused on clean, maintainable code through OOP, SOLID principles, and design patterns including State Machine, Observer, MVC, and Dependency Injection. Background includes shipped mobile games on iOS and Android, software validated in a clinical RCT with 100 patients, and co-authored publications in Nature Scientific Reports and Frontiers in Virtual Reality.

EXPERIENCE

Unity Developer · IDIBAPS April 2021 – Present

Barcelona, Spain · Institut d'Investigacions Biomèdiques August Pi i Sunyer

- Developed and maintained 10+ Unity applications for interactive simulation and clinical research, delivered as part of EU-funded projects (VR-PerGenere, XR-Pain / EMIL XR).
- Designed and led the Unity architecture of XR-Pain alongside one collaborator — a real-time interactive application validated in a Randomised Controlled Trial (RCT) with 100 patients.
- Defined codebase standards across projects using State Machine, Observer/Events, MVC, Object Pooling, Scriptable Objects, and Dependency Injection — ensuring modularity, testability, and long-term maintainability.
- Built avatar and embodiment systems: real-time avatar control, inverse kinematics, blend shape animation, and body ownership illusion mechanics, driving the core research hypotheses.
- Collaborated closely with researchers, clinicians, and designers in an interdisciplinary team, translating scientific requirements into robust, reusable Unity systems.

Unity Developer · EventLab (Universitat de Barcelona) Sept 2019 – June 2020

Barcelona, Spain

- Designed and built an interactive Unity application for a research study on vibrotactile feedback and user perception, delivered from concept through a 10-participant pilot.
- Implemented cross-platform builds from a shared codebase targeting multiple hardware configurations.

SKILLS

Engine: Unity 2019–2023 LTS, C#, Built-in Render Pipeline, URP (familiar)

Patterns: State Machine, Events, MVC, Object Pooling, Scriptable Objects, Dependency Injection, Singleton

Systems: Avatar & animation systems, IK, physics, mobile input, real-time interaction, UI (uGUI)

Tools: Visual Studio, Rider, Git, 3ds Max, Trello, Slack, Unity Profiler

XR: Meta XR SDK, OpenXR, Unity XR Interaction Toolkit, Hand Tracking, Meta Quest 2/3

Other: C++, Python, Agile/Scrum

PERSONAL PROJECTS

Balanzi! — Mobile Puzzle Game (iOS & Android) [App Store](#) · [Google Play](#) June 2025 – Present

Kelgar Interactive · 2-person indie team

- Co-developing a physics-based mobile puzzle game in Unity, live on both iOS and Android, in which players stack and balance geometric shapes on a moving platform against the clock.
- Responsible for gameplay systems design and implementation: physics interactions, difficulty scaling, scoring system, global leaderboard integration, and mobile touch controls.
- Delivered multiple post-launch updates, including tutorial system, pause/resume flow, per-shape scoring, and UI improvements — maintaining quality on both platforms simultaneously.

Koala Games — 3D Art & Unity Integration [instagram/koalagamesdev](https://www.instagram.com/koalagamesdev) Sept 2019 – July 2020

- Created all character and environment 3D assets for two games (Olymprix, GetApp) using 3ds Max: modelling, UV mapping, texturing, rigging, and skinning.
- Programmed Unity integration tooling and a runtime character customisation system to manage and assemble modular assets within the engine.

VR Puzzle Game — Hand Tracking & OptiTrack [github/edirc98/VirVigInteractionGame](https://github.com/edirc98/VirVigInteractionGame) 2020

- Developed a Unity puzzle game built around hand tracking interactions (grabbing, throwing, placement), integrated with an OptiTrack motion-capture suit to record upper-body movement for a motion-reconstruction AI dataset.

EDUCATION

MSc Innovation and Research in Informatics (MIRI) 2021 – 2024

Universitat Politècnica de Catalunya (UPC)

Specialisation: Computer Graphics and Virtual Reality

BA Interactive Digital Contents 2016 – 2020

Universitat de Barcelona – ENTI UB

Focus: Video game design, graphics programming, agile methodologies

PUBLICATIONS

Świdrak J., Arias A., Rodríguez de la Calle E., Collado Cruz A., Sanchez-Vives M.V. *Virtual embodiment in fibromyalgia*. Scientific Reports (Nature), 13:10719, 2023. [doi:10.1038/s41598-023-36861-3](https://doi.org/10.1038/s41598-023-36861-3)

Amestoy Alonso et al. (incl. Rodríguez de la Calle E.). *Focus groups in the metaverse: shared virtual spaces for patients, clinicians, and researchers*. Frontiers in Virtual Reality, 5, 2024. [doi:10.3389/frvir.2024.1432282](https://doi.org/10.3389/frvir.2024.1432282)

LANGUAGES

Spanish — Native **Catalan** — Native **English** — Professional working proficiency (B2 certified; daily use in research environment) **French** — Basic